电子科技大学 格拉斯哥 学院

UOG-UESTC Joint School of UESTC

标 准 实 验 报 告

Lab Report

（实验）课程名称： 信号与系统

(LAB) Course Name： Signals and Systems

电子科技大学教务处制表

UoG-UESTC Joint School
Lab Report

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Instructor：Aynul Islam

Location：No.6 Scientific Research Build **Date：2024.9.23**

一、 Laboratory name：

Signals and Systems

二、 Project name：

Represent signals using MATLAB

三、 Lab hours：4

四、 Theoretical background：

1. The basic concepts of signals and systems arise in a variety of contexts, from engineering design to financial analysis. In this lab1, you will learn how to represent, manipulate, and analyze basic signals and systems in MATLAB.

2. Some basic MATLAB commands for representing signals include: zeros, ones, cos, sin, exp, real, imag, abs, angle, linspace, plot, stem, subplot, xlabel, ylabel, title

3. Some useful commands in Symbolic Math Toolbox are as: sym, subs, ezplot.

五、 Objective：

1. Familiar with some basic MATLAB commands to represent and plot continuous-time and discrete-time signals.
2. Perform operations, including transformation, on signals using MATLAB.
3. Analyze period of signals using MATLAB.
4. Calculate energy and power of signals using MATLAB.

六、 Description：

The following exercises are from the book," John R.Buck, Michael M. Daniel, Andrew C. Singer. Computer Exploration in Signals and Systems —— Using MATLAB."

1. Plot the discrete time signals and determine the period. 1.2(d)
2. Perform the transformation of signals. 1.3 (a)(b)(c)
3. Analyze the properties of systems. 1.4(a)(b)(c)(f)
4. Plot signals using Symbolic Math toolbox. 1.6(a)
5. Calculate energy and power of signals. 1.8(a)(b)(c)(d)(e)(f)

七、 Required instruments：

Computer, MATLAB

八、 Procedures, Analysis of Lab data & result and Conclusion：

1．Plot the discrete time signals and determine the period.

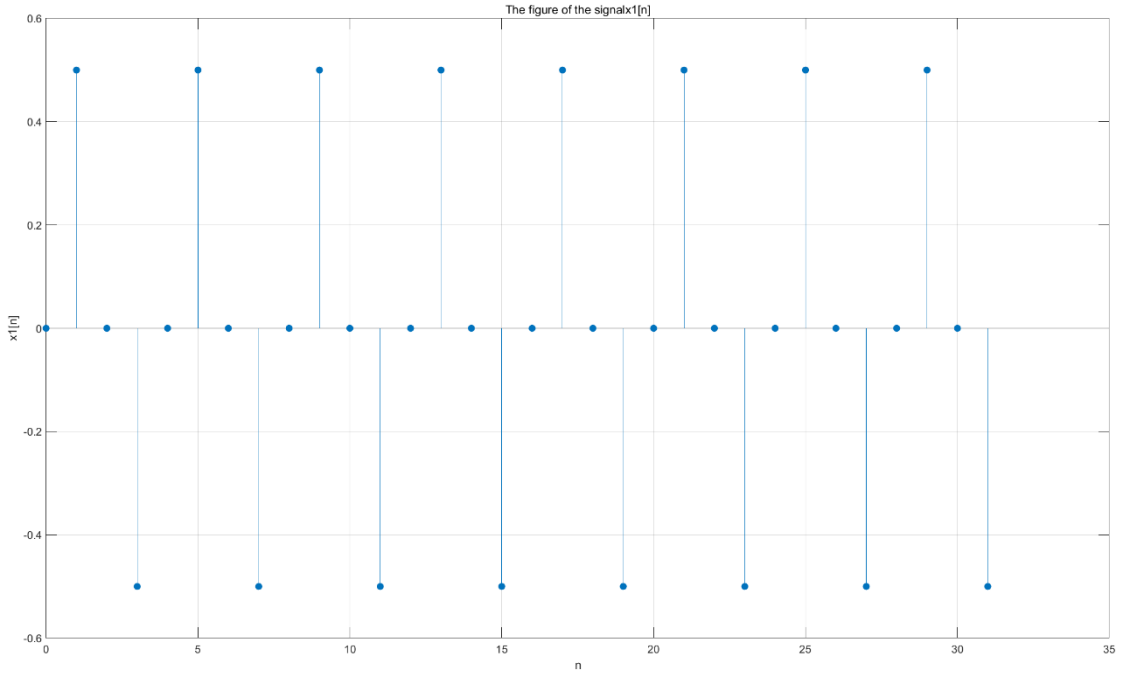
1.2

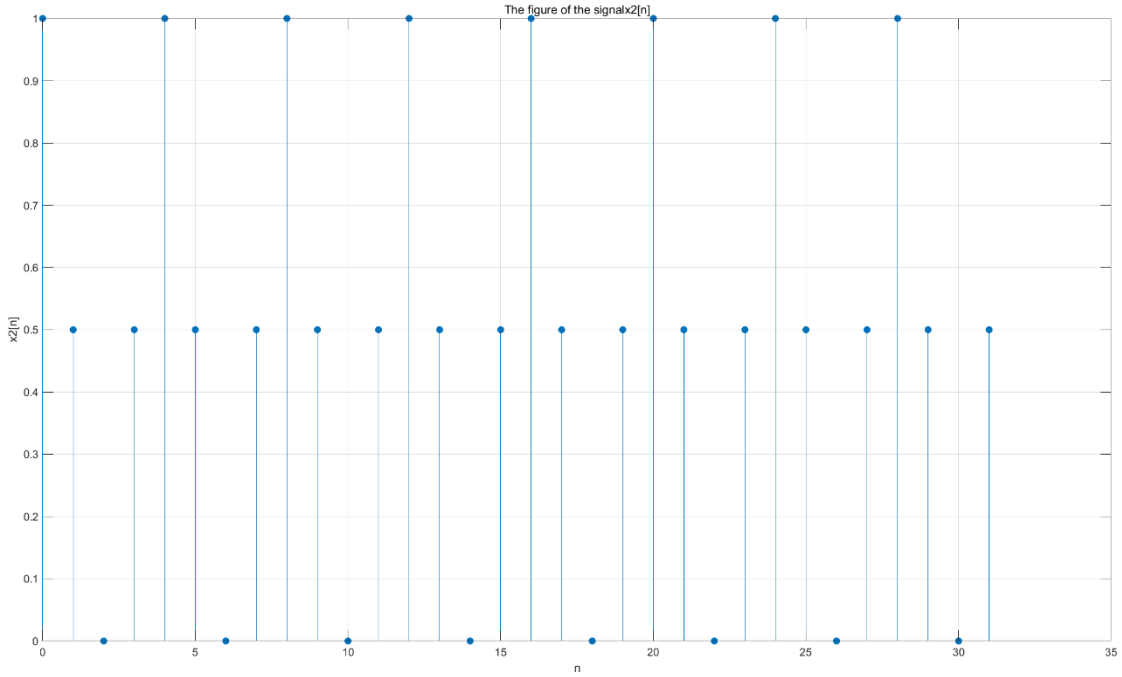
(d)

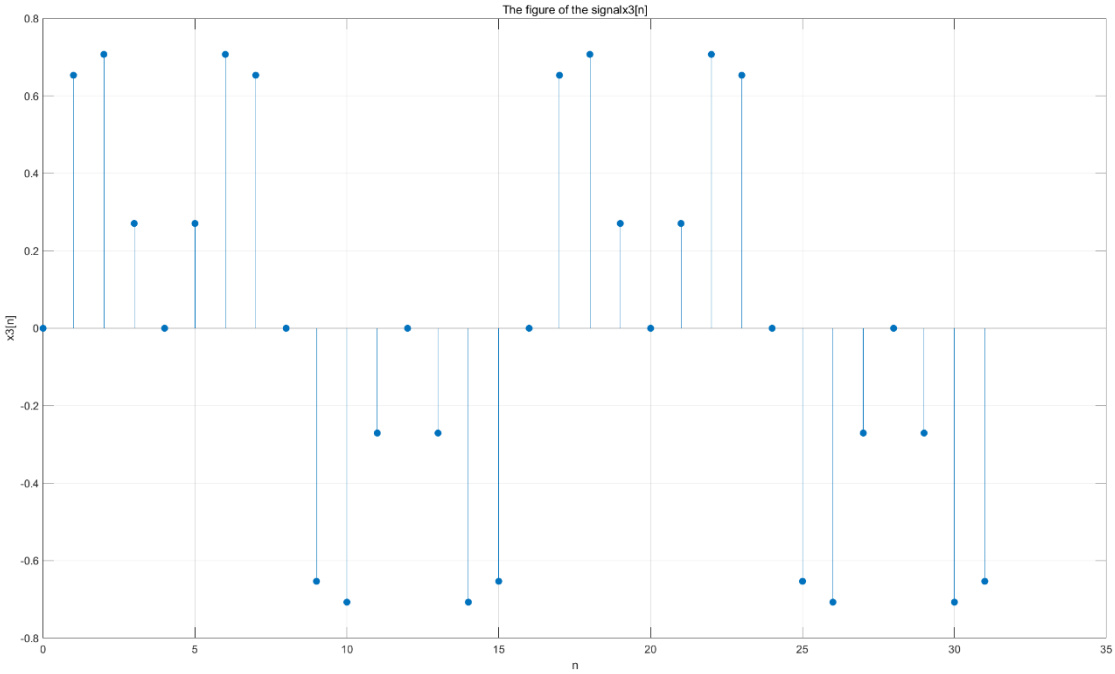
Codes:

```
n=0:31;
x1=sin(pi/4*n).*cos(pi/4*n);
figure(1);
stem(n,x1,'fill');
xlabel('n');
ylabel('x1[n]');
title('The figure of the signalx1[n]');
grid on;
x2=cos(pi/4*n).^2;
figure(2);
stem(n,x2,'fill');
xlabel('n');
ylabel('x2[n]');
title('The figure of the signalx2[n]');
grid on;
x3=sin(pi/4*n).*cos(pi/8*n);
figure(3);
stem(n,x3,'fill');
xlabel('n');
ylabel('x3[n]');
title('The figure of the signalx3[n]');
grid on;
```

Figure:


The figure of the signalx1[n]
x1[n]
n
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
0
5
10
15
20
25
30
35


The figure of the signalx2[n]
x2[n]
n
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0
5
10
15
20
25
30
35



Explanations:

$$x_1[n] = \sin(\frac{\pi n}{4})\cos(\frac{\pi n}{4}) = \frac{1}{2}\sin(\frac{\pi n}{2})$$

$$x_2[n] = \cos^2(\frac{\pi n}{4}) = \frac{1}{2}[\cos(\frac{\pi n}{2}) + 1]$$

$$x_3[n] = \sin(\frac{\pi n}{4})\cos(\frac{\pi n}{8})$$

According to T=

$\left|\frac{2\pi}{\omega}\right|$, the fundamental period of the signal x_1 [n] x_2[n] x_3 [n]is 4,4,16,.

Which is corresponding to the figures generated by matlab.

2. Perform the transformation of signals.

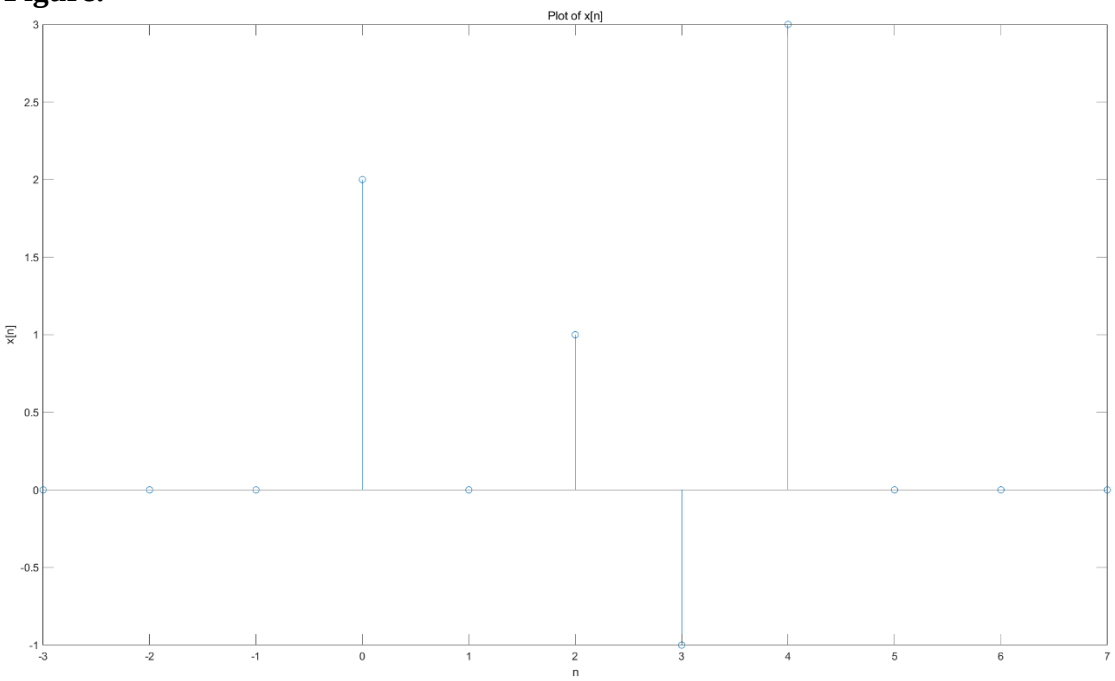
1.3

(a)

Codes:

```
clf;
nx=[-3:7];
x=[zeros(1,3) 2 0 1 -1 3 zeros(1,3)];
figure(1);
stem(nx,x);
xlabel ('n');ylabel('x[n]');title('Plot of x[n]');
```

Figure:

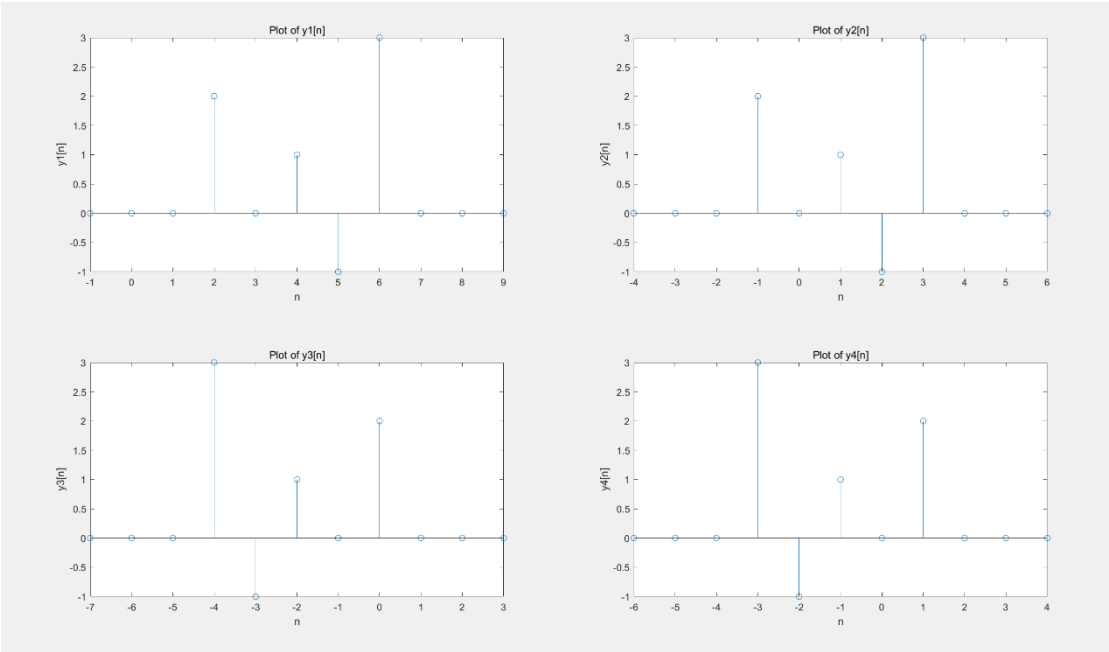


b and c

codes:

```
ny1=nx+2;
y1=x;
subplot(2,2,1);
stem(ny1,y1);
xlabel ('n');ylabel('y1[n]');title('Plot of y1[n]');
ny2=nx-1;
y2=x;
subplot(2,2,2);
stem(ny2,y2);
xlabel ('n');ylabel('y2[n]');title('Plot of y2[n]');
ny3=-nx;
y3=x;
subplot(2,2,3);
stem(ny3,y3);
xlabel ('n');ylabel('y3[n]');title('Plot of y3[n]');
ny4=-nx+1;
y4=x;
subplot(2,2,4);
stem(ny4,y4);
xlabel ('n');ylabel('y4[n]');title('Plot of y4[n]');
```

Figure



Explanations

The figure of x[n] is shown in 1.3(a), The figure of$y_1[n]$ $y_2[n]$ $y_3[n]$ $y_4[n]$is shown in 1.3(a)， according to these figures:

$y_1[n]$ is $x[n]$delayed by 2.

$y_2[n]$ is $x[n]$advanced by 1.

$y_3[n]$ is $x[n]$flipped.

$y_4[n]$ is $x[n]$advanced by 1 and then flipped.

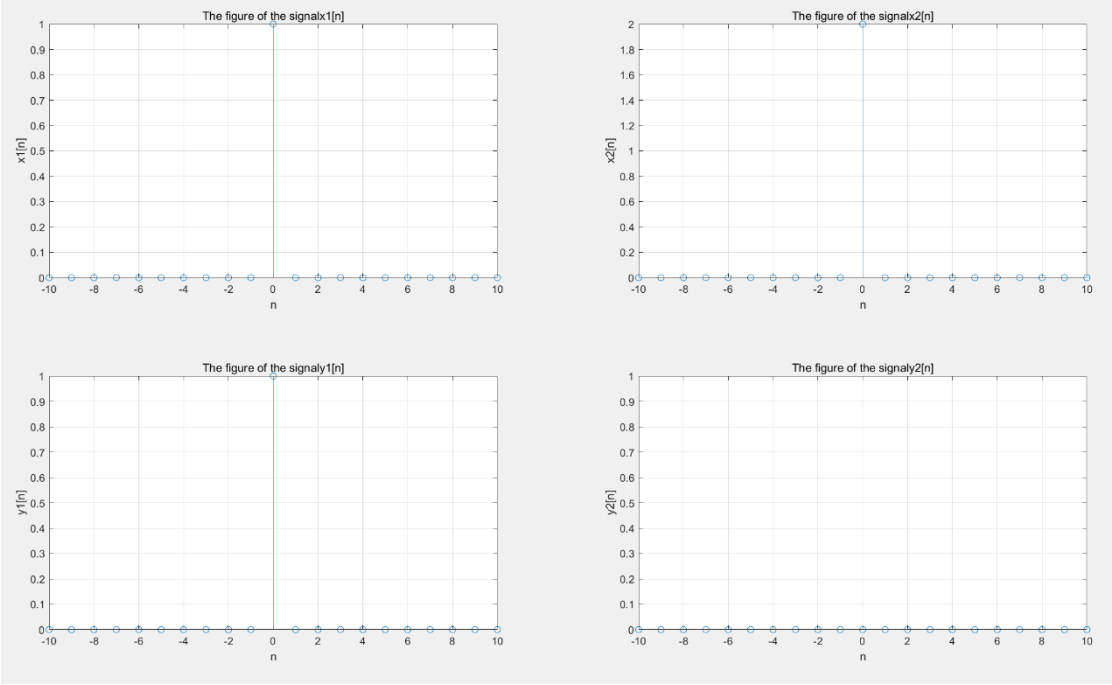
3. Analyze the properties of systems.

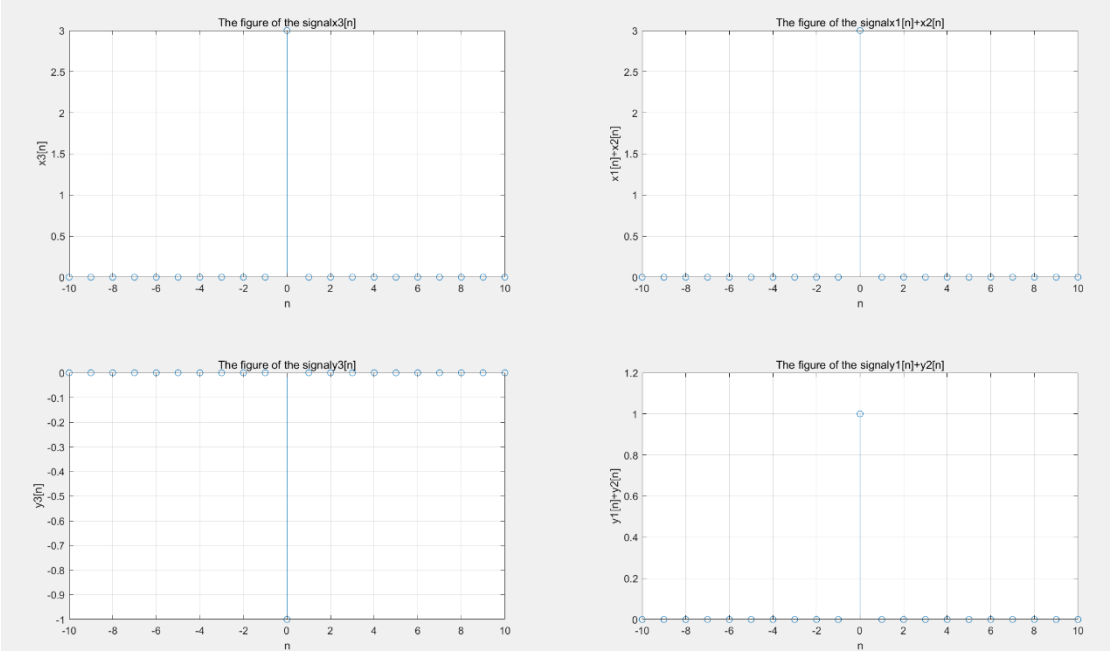
1.4(a)(b)(c)(f)

a)

```
  n=[-10:10];
x1=[zeros(1,10) 1 zeros(1,10)];
x2=[zeros(1,10) 2 zeros(1,10)];
y1=sin(pi/2*x1);
y2=sin(pi/2*x2);
figure(1);
subplot(2,2,1);
```

```
stem(n,x1),xlabel('n');ylabel('x1[n]');title('The figure of the signalx1[n]');
grid on;
subplot(2,2,2);
stem(n,x2);xlabel('n');ylabel('x2[n]');title('The figure of the signalx2[n]');
grid on;
subplot(2,2,3);
stem(n,y1);xlabel('n');ylabel('y1[n]');title('The figure of the signaly1[n]');
grid on;
subplot(2,2,4);
stem(n,y2);xlabel('n');ylabel('y2[n]');title('The figure of the signaly2[n]');
grid on;
axis([-10,10,0,1]);
x3=x1+x2;
y3=sin(pi/2*x3);
figure(2);
subplot(2,2,1);
stem(n,x3);xlabel('n');ylabel('x3[n]');title('The figure of the signalx3[n]');
grid on;
subplot(2,2,2);
stem(n,x1+x2);xlabel('n');ylabel('x1[n]+x2[n]');title('The figure of the
signalx1[n]+x2[n]');
grid on;
subplot(2,2,3);
stem(n,y3);xlabel('n');ylabel('y3[n]');title('The figure of the signaly3[n]');
grid on;
subplot(2,2,4);
stem(n,y1+y2);xlabel('n');ylabel('y1[n]+y2[n]');title('The figure of the
signaly1[n]+y2[n]');
grid on;
figure
```





Explanation:

From the figure, we can get that $x_1[n] = 2x_2[n]$, but $y_1[n] \neq 2y_2[n]$.

And, $x_3[n] = x_1[n] + x_2[n]$, but $y_3[n] \neq y_1[n] + y_2[n]$.

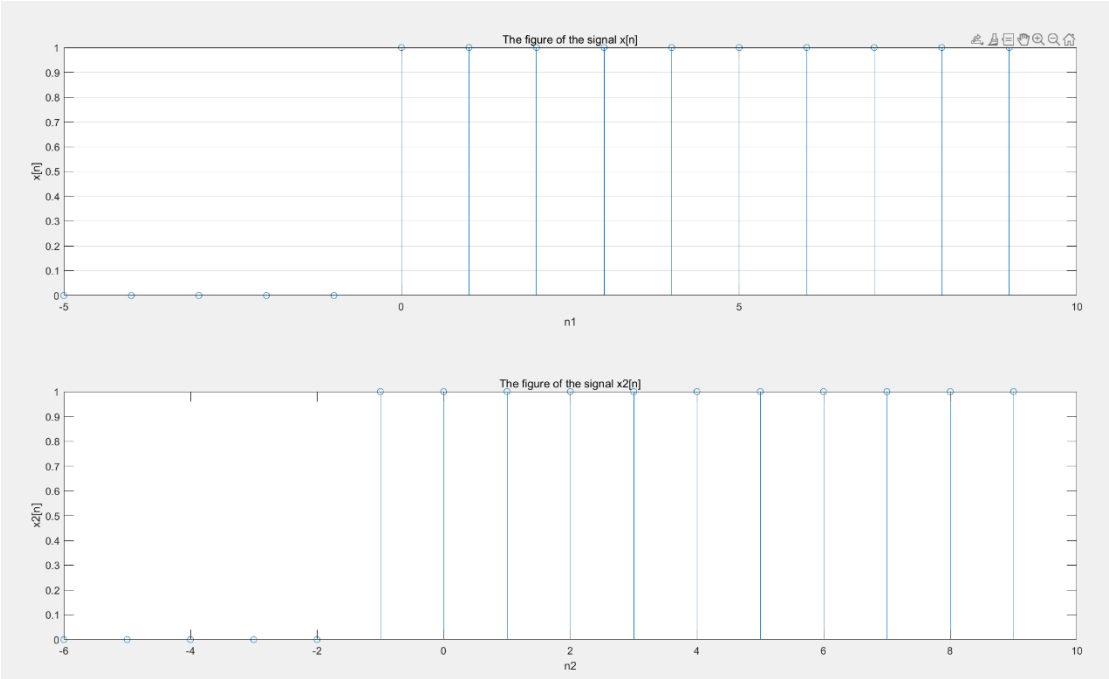
Therefore, the system is not linear.

b)

codes:

```
n1=-5:9;
x=[zeros(1,5) ones(1,10)];
figure(1)
subplot(2,1,1);
stem(n1,x);xlabel('n1');ylabel('x[n]');title('The figure of the signal x[n]');
grid on;
n2=-6:9;
x1=[zeros(1,6) ones(1,10)];
x2=[zeros(1,5) ones(1,11)];
subplot(2,1,2);
stem(n2,x2);xlabel('n2');ylabel('x2[n]');title('The figure of the signal x2[n]');
y=x1+x2;
figure(2);
stem(n2,y);xlabel('n2');ylabel('y[n]');title('The figure of the signal y[n]');
grid on;
```

figure



Explanation:

y[n]=x[n]+x[n+1], from the figure and equation, we can get that the output of the system depends on the future values of the input. Thus, the system is not causal.

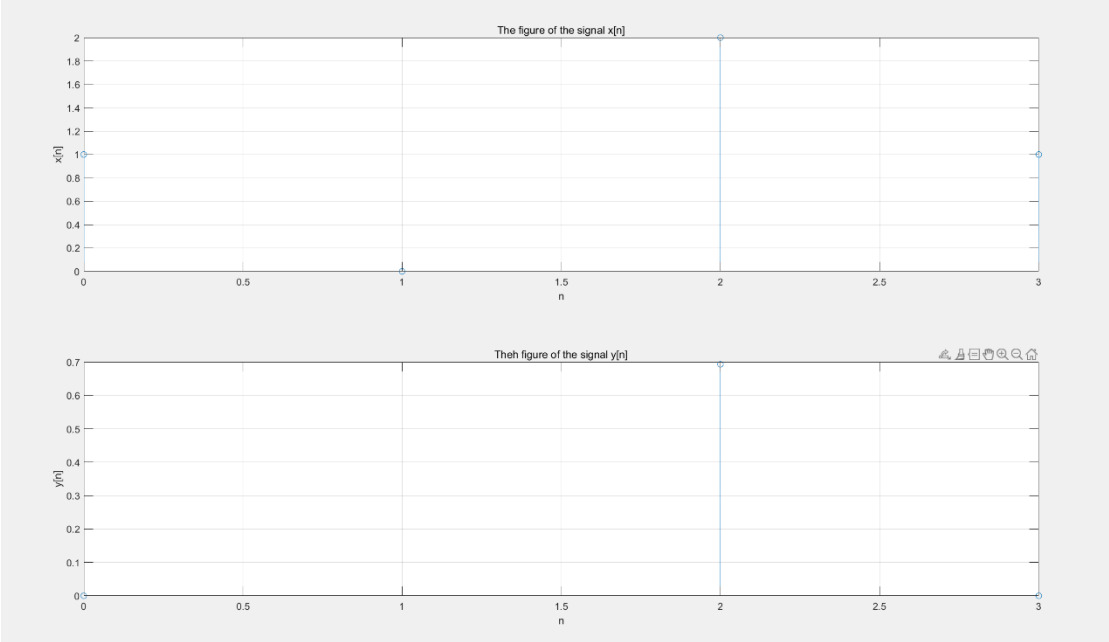
c)

code:

```
clc
n=[0:3];
x=[1 0 2 1];
y=log(x);
```

```
subplot(2,1,1);
stem(n,x);xlabel('n');ylabel('x[n]');title('The figure of the signal x[n]');
grid on;
subplot(2,1,2);
stem(n,y);xlabel('n');ylabel('y[n]');title('Theh figure of the signal y[n]');
grid on;
```

figure:



Explanation:

When x[1]=0,the value of y[1] does not exist.This indicates when the input x is finite, the output y is infinite,which means the system is unstable.

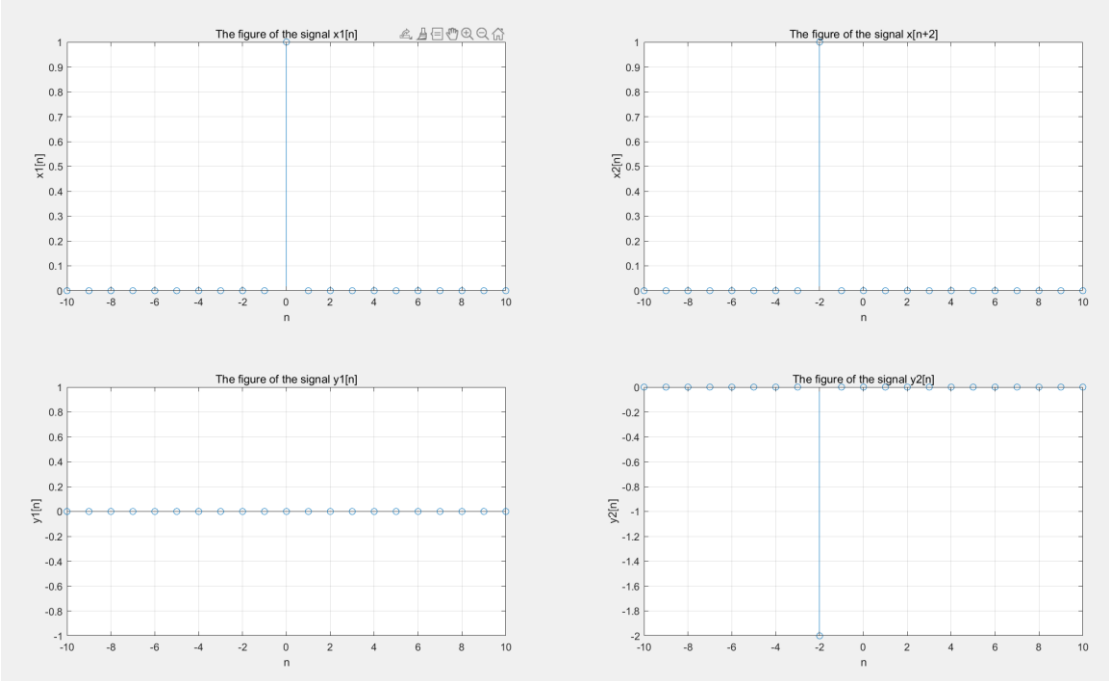
f)code:

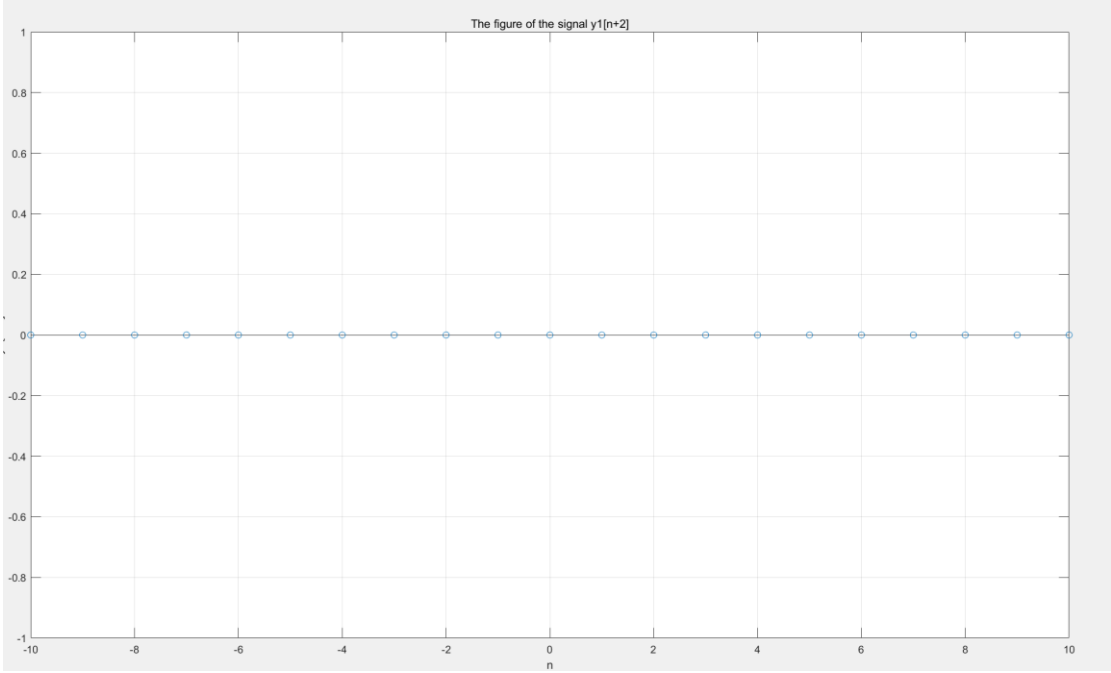
1.1.4.f_1 Prove the system is not time-invariant

```
n=[-10:10];
x1=[zeros(1,10) 1 zeros(1,10)];
x2=[zeros(1,8) 1 zeros(1,12)];
figure(1);
subplot(2,2,1);
stem(n,x1);xlabel('n');ylabel('x1[n]');title('The figure of the signal x1[n]');
grid on;
subplot(2,2,2);
stem(n,x2);xlabel('n');ylabel('x2[n]');title('The figure of the signal x[n+2]');
grid on;
y1=n.*x1;
subplot(2,2,3);
stem(n,y1);xlabel('n');ylabel('y1[n]');title('The figure of the signal y1[n]');
grid on;
y2=n.*x2;
```

```
subplot(2,2,4);
stem(n,y2);xlabel('n');ylabel('y2[n]');title('The figure of the signal y2[n]');
grid on;
y3=(n+2).*x2;
figure(2);
stem(n,y3);xlabel('n');ylabel('y1[n+2]');title('The figure of the signal y1[n+2]');
grid on;
```

figure





Explanation:

X1[N+2]=X2[N+2]

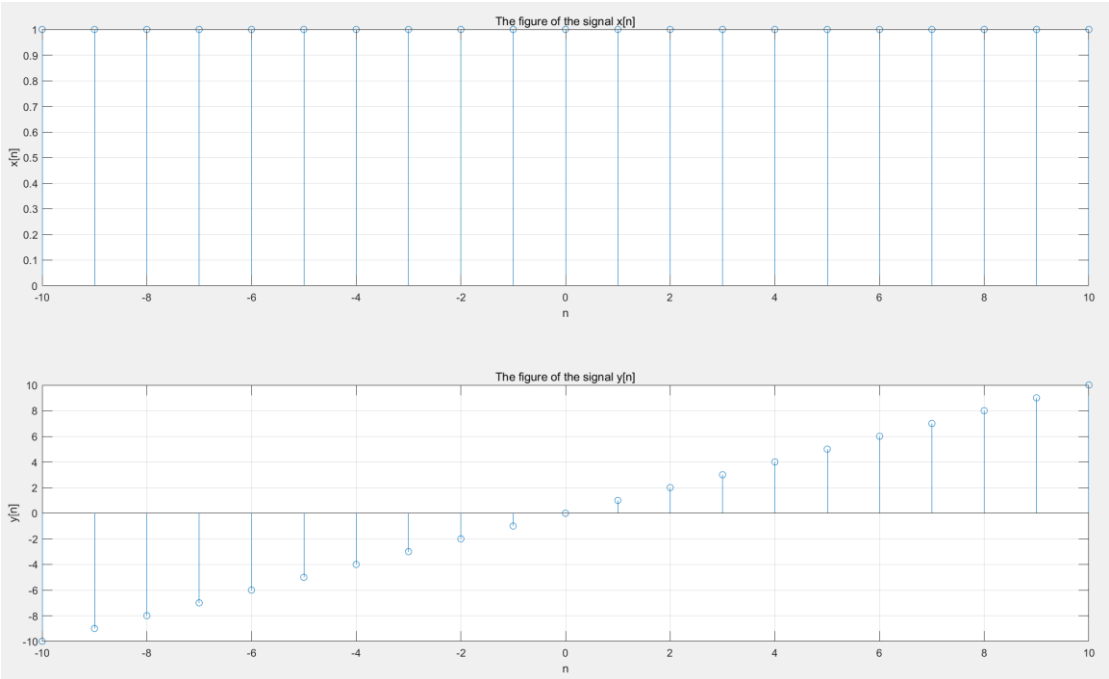
Y1[N]=N*X1[N]
Y2[N]=N*X2[N]
Y3[N]=Y1[N+2]=(N+2)*X1[N+2]
From the figures,Y1[N+2]≠Y2[N]. Therefore, a time shift in the input signal does not result in an identical time shift in the output signal, and thus the system is not time-invariant.

1.4.f.(2) Prove the system is not stable:

Code:

```
n=[-10:10];
x=[ones(1,21)];
subplot(2,1,1);
stem(n,x);xlabel('n');ylabel('x[n]');title('The figure of the signal x[n]');
grid on;
y=n.*x;
subplot(2,1,2);
stem(n,y);xlabel('n');ylabel('y[n]');title('The figure of the signal y[n]');
grid on;
```

figure:



Explanation:

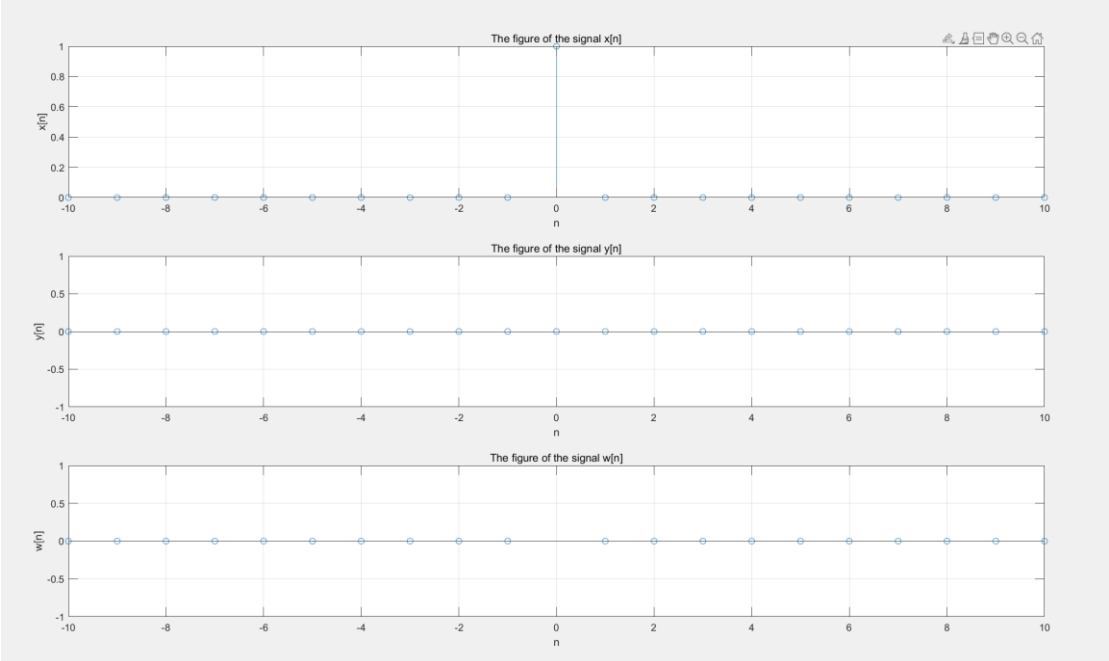
From the figure, it's clear that while the input remains bounded, the output is unbounded. As n approaches infinity, the output diverges, indicating that the system is unstable.

1.4.f. (3) Prove the system is not invertible:

Code:

```
n=[-10:10];
x=[zeros(1,10) 1 zeros(1,10)];
y=n.*x;
subplot(3,1,1);
stem(n,x);xlabel('n');ylabel('x[n]');title('The figure of the signal x[n]');
grid on;
subplot(3,1,2);
stem(n,y);xlabel('n');ylabel('y[n]');title('The figure of the signal y[n]');
grid on;
w=1./n.*y;
subplot(3,1,3);
stem(n,w);xlabel('n');ylabel('w[n]');title('The figure of the signal w[n]');
grid on;
```

figure:



Explanation:

If the system were invertible, the figure of X[N] and W[N] would be identical. However, as shown in the figure, it is clear that they are ddifferent. Thus, the system is not invertible.

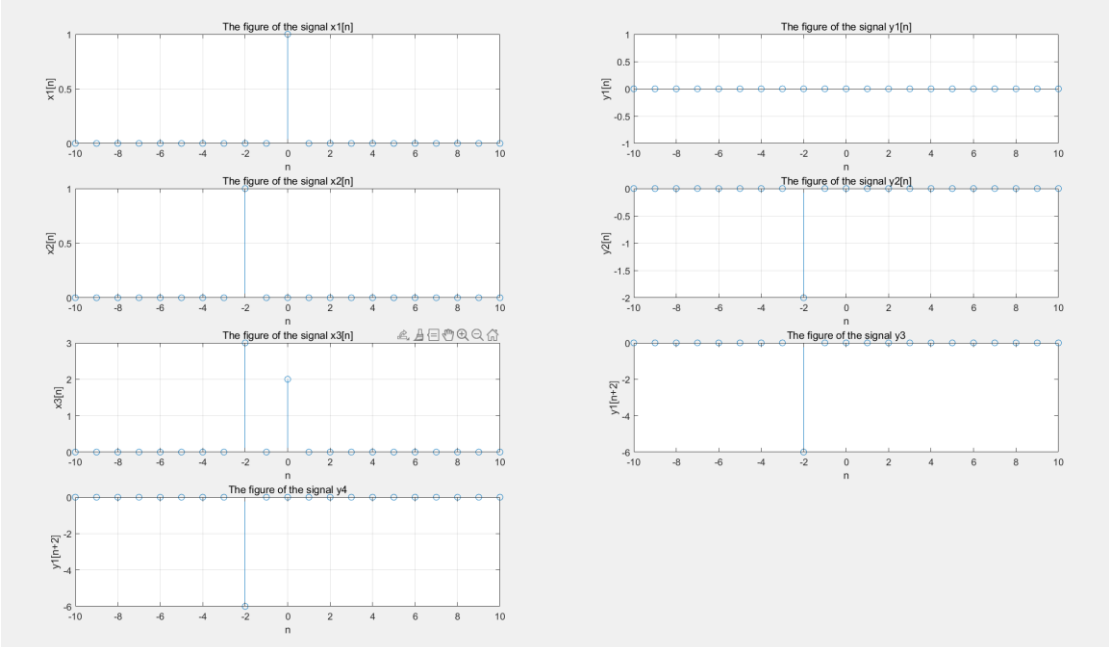
1.4.f.(4) Prove the system is linear:

Code:

```
n=[-10:10];
x1=[zeros(1,10) 1 zeros(1,10)];
x2=[zeros(1,8) 1 zeros(1,12)];
subplot(4,2,1);
```

```
stem(n,x1);xlabel('n');ylabel('x1[n]');title('The figure of the signal x1[n]');
grid on;
y1=n.*x1;
subplot(4,2,2);
stem(n,y1);xlabel('n');ylabel('y1[n]');title('The figure of the signal y1[n]');
grid on;
subplot(4,2,3);
stem(n,x2);xlabel('n');ylabel('x2[n]');title('The figure of the signal x2[n]');
grid on;
y2=n.*x2;
subplot(4,2,4);
stem(n,y2);xlabel('n');ylabel('y2[n]');title('The figure of the signal y2[n]');
grid on;
x3=2*x1+3*x2;
subplot(4,2,5);
stem(n,x3);xlabel('n');ylabel('x3[n]');title('The figure of the signal x3[n]');
grid on;
y3=n.*x3;
subplot(4,2,6);
stem(n,y3);xlabel('n');ylabel('y1[n+2]');title('The figure of the signal y3');
y4=2*y1+3*y2;
subplot(4,2,7);
stem(n,y4);xlabel('n');ylabel('y1[n+2]');title('The figure of the signal y4');
grid on;
```

figure:



Explanation:

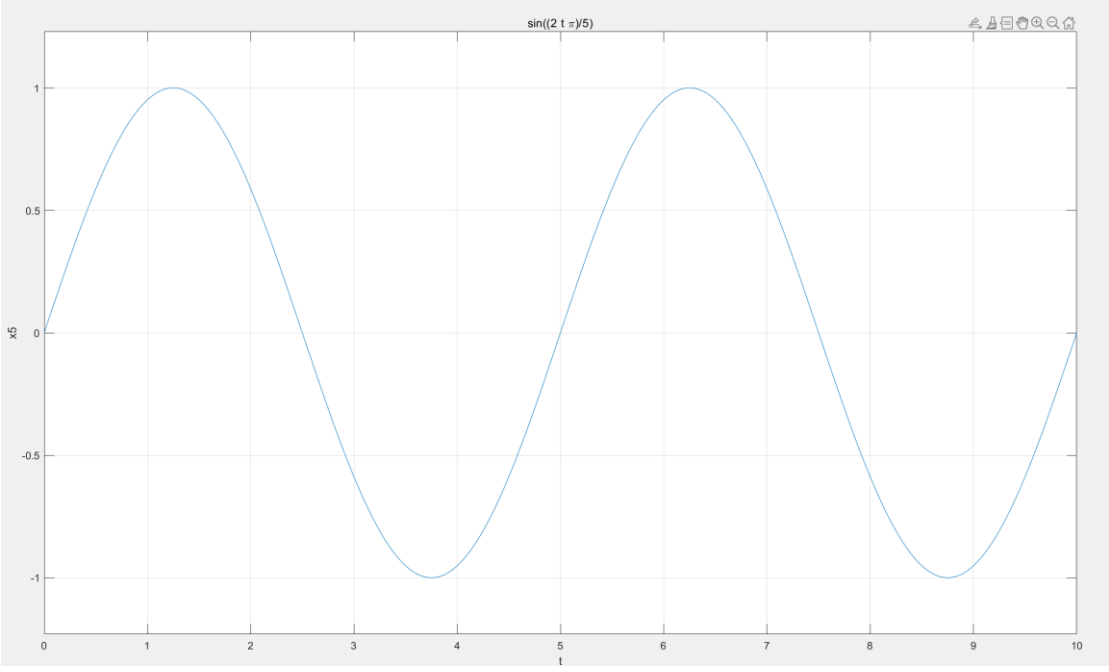
X3[N]=2*X1[N]+3*X2[N]
Y1[N]=N*X1[N]
Y2[N]=N*X2[N]
Y3[N]=N*X3[N]
Y4[N]=2*Y1[N]+3*Y2[N]
From the figure, we can get that X3[N]=2*X1[N]+3*X2[N] , Y4[N]=2*Y1[N]+3*Y2[N]=Y3[N]. Therefore, the system is linear.

1.6.a

Code:

```
clc
syms T t
x = sin(2*pi*t/T);
x5 = subs(x, T, 5);
ezplot(x5, [0, 10]);
xlabel('t');
ylabel('x5');
grid on;
```

figure:



Explanation

The function ezplot can be used to plot a symbolic expression which has only one variable.

1.8.ab

Code:

```
clc
clear
x1 = str2sym('cos(pi*t/5)');
x2 = str2sym('sin(pi*t/5)');
m = str2sym('exp(i*2*pi*t/3)');
n = str2sym('exp(i*pi*t)');
x3 = m + n;

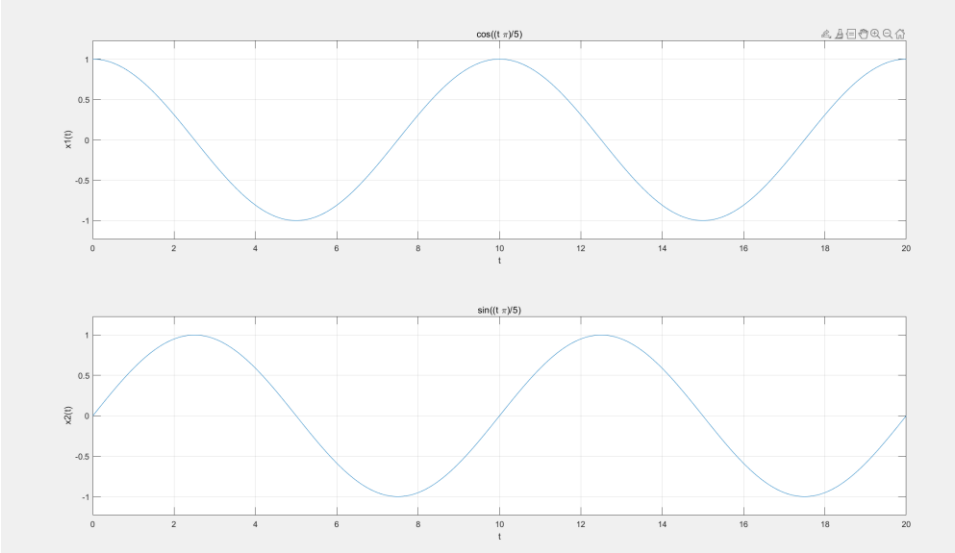
figure(1);
subplot(2,1,1);
ezplot(x1, [0,20]);
xlabel('t');
ylabel('x1(t)');
grid on;

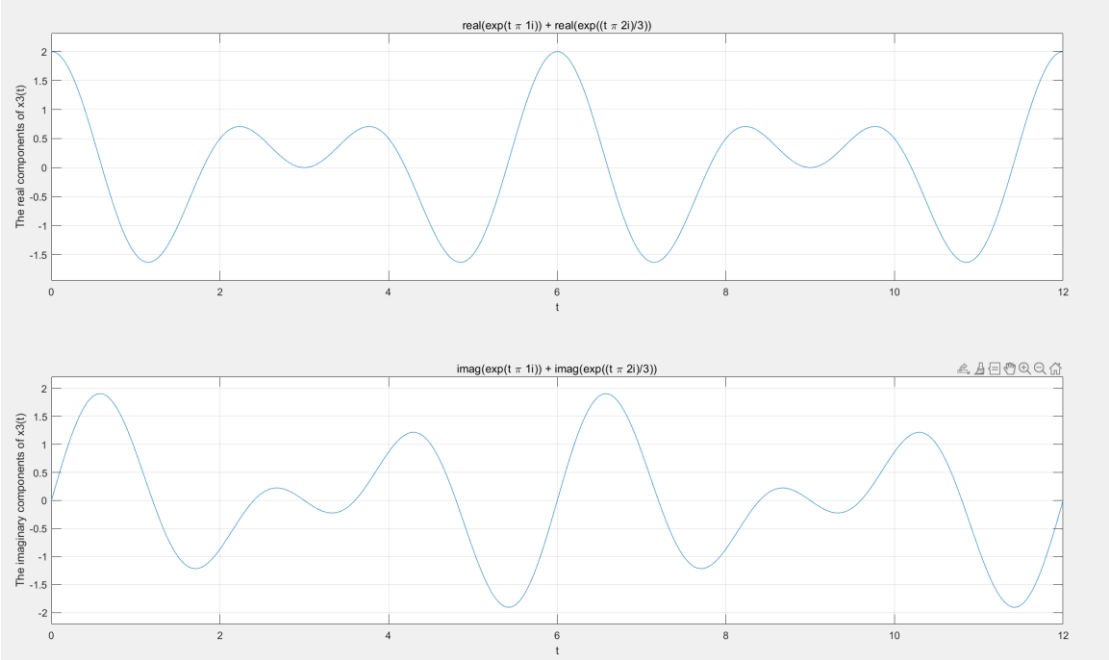
subplot(2,1,2);
ezplot(x2, [0,20]);
xlabel('t');
ylabel('x2(t)');
grid on;

figure(2);
subplot(2,1,1);
ezplot(real(x3), [0,12]);
xlabel('t');
ylabel('The real components of x3(t)');
grid on;

subplot(2,1,2);
ezplot(imag(x3), [0,12]);
xlabel('t');
ylabel('The imaginary components of x3(t)');
grid on;
```

figure





1.8.cd

Code:

```
 clc
clear
x1 = str2sym('cos(pi*t/5)');
x2 = str2sym('sin(pi*t/5)');
x3 = str2sym('exp(i*2*pi*t/3) + exp(i*pi*t)');

syms a t real;
E1 = int(x1^2, [-a, a]);
E2 = int(x2^2, [-a, a]);
```

```
x3c = conj(x3);
E3 = int(x3*x3c, [-a, a]);

figure(1);
subplot(3,1,1);
ezplot(E1, [0, 30]);
xlabel('a');
ylabel('Ea for x1');
title('Plot of Ea for x1');

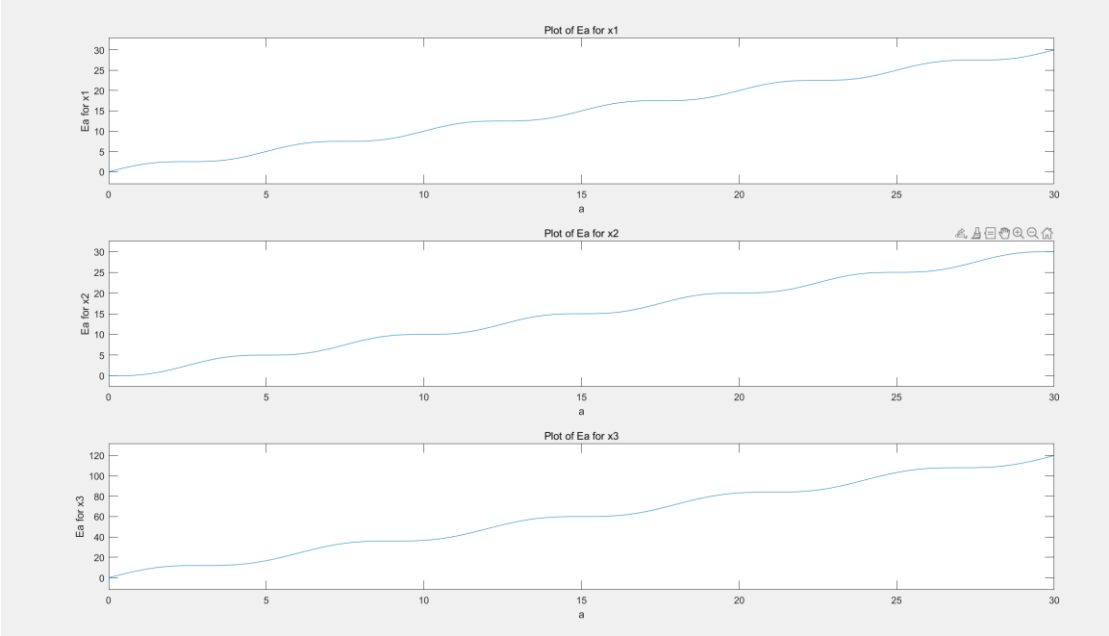
subplot(3,1,2);
ezplot(E2, [0, 30]);
xlabel('a');
ylabel('Ea for x2');
title('Plot of Ea for x2');

subplot(3,1,3);
ezplot(E3, [0, 30]);
xlabel('a');
ylabel('Ea for x3');
title('Plot of Ea for x3');

Et1 = int(x1^2, [-5, 5]);
Et2 = int(x2^2, [-5, 5]);
ET3 = int(x3*x3c, [-3, 3]);

Et3 = eval(ET3);
disp(Et1);
disp(Et2);
disp(Et3);
```

figure:



Explanation:

The $E_{T/2}$ of $x_1(t)$ is 5, the $E_{T/2}$ of $x_2(t)$ is 5, the $E_{T/2}$ of $x_3(t)$ is 12. The energy increases as the interval length increases. The expected values for E_∞ is infinities.

1.8.ef

Code:

```
clc
clear
x1 = str2sym('cos(pi*t/5)');
x2 = str2sym('sin(pi*t/5)');
x3 = str2sym('exp(i*2*pi*t/3) + exp(i*pi*t)');

syms a t real;
E1 = int(x1^2, [-a, a]);
E2 = int(x2^2, [-a, a]);

x3c = conj(x3);
E3 = int(x3*x3c, [-a, a]);

P1 = E1 / (2 * a);
P2 = E2 / (2 * a);
P3 = E3 / (2 * a);

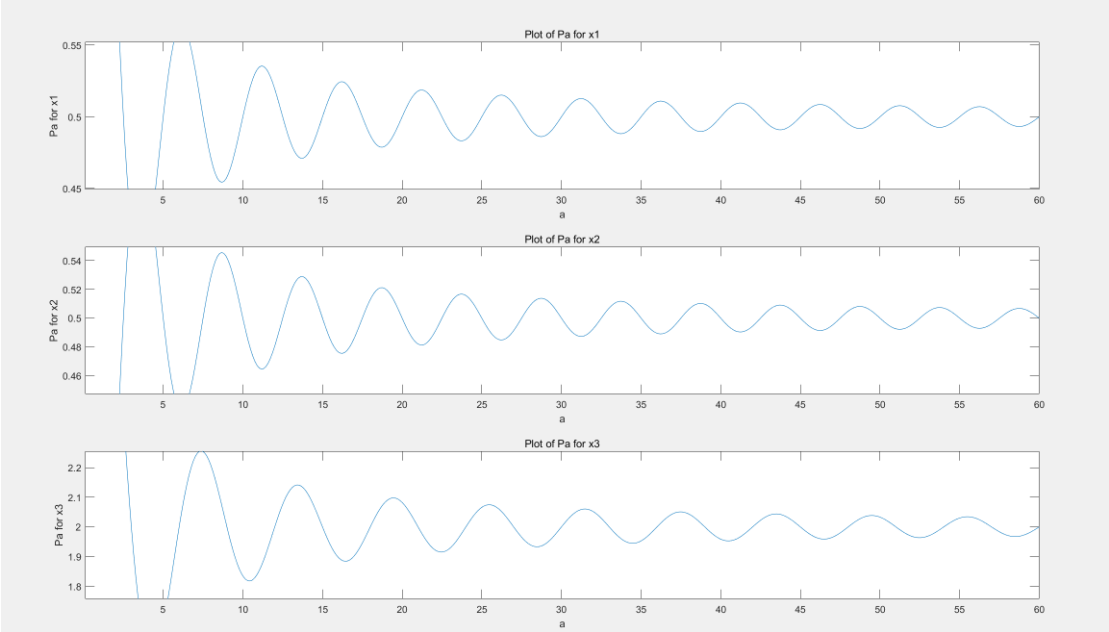
figure(1);
```

```
subplot(3,1,1);
ezplot(P1, [0.1, 60]);
xlabel('a');
ylabel('Pa for x1');
title('Plot of Pa for x1');

subplot(3,1,2);
ezplot(P2, [0.1, 60]);
xlabel('a');
ylabel('Pa for x2');
title('Plot of Pa for x2');

subplot(3,1,3);
ezplot(P3, [0.1, 60]);
xlabel('a');
ylabel('Pa for x3');
title('Plot of Pa for x3');
sumP = P1 + P2;
r = simplify(sumP);
disp('Simplified sum of P1 and P2:');
disp(r);
```

figure:



Explanation:

1） P_a varies and tends to be a finite value as a increases. And we already get the $E_{T/2}$ of $x_1(t)$, $x_2(t)$ and $x_3(t)$ from (d).

For $x_1(t)$, the expected value for P_∞ is 0.5,and it's $\frac{E_{T/2}}{T}=\frac{5}{10}=0.5$.Therefore, for $x_1(t)$, $P_\infty=\frac{E_{T/2}}{T}$.

For $x_2(t)$, the expected value for P_∞ is 0.5,and it's $\frac{E_{T/2}}{T}=\frac{5}{10}=0.5$.Therefore, for $x_2(t)$, $P_\infty=\frac{E_{T/2}}{T}$.

For $x_3(t)$, the expected value for P_∞ is 2,and it's $\frac{E_{T/2}}{T}=\frac{12}{6}=2$.Therefore, for $x_3(t)$, $P_\infty=\frac{E_{T/2}}{T}$.

$P_\infty=\lim_{a\to\infty}P_a$, which is the time-average power of the entire signal. Since $E_{T/2}$ is the energy in a single period of the signal and the duration of time is T, so $\frac{E_{T/2}}{T}$ can be also on behalf of the time-average power. Therefore, $P_\infty=\frac{E_{T/2}}{T}$ for each given signal can be anticipated.

(1) The answer of (f):

For $x_1(t)$, $P_1=\frac{\int_{-a}^{a}|\cos(\frac{\pi t}{5})|^2 dt}{2a}=\frac{5}{4a\pi}\sin(\frac{2\pi a}{5})+\frac{1}{2}$, $P_\infty=\lim_{a\to\infty}P_a=\frac{1}{2}$.

For $x_2(t)$, $P_2=\frac{\int_{-a}^{a}|\sin(\frac{\pi t}{5})|^2 dt}{2a}=\frac{1}{2}-\frac{5}{4a\pi}\sin(\frac{2\pi a}{5})$, $P_\infty=\lim_{a\to\infty}P_a=\frac{1}{2}$

Therefore, we anticipate that Pa for x_1(t) and Pa for x_2(t) will converge to the same value..

$P_1+P_2=\frac{5}{4a\pi}\sin(\frac{2\pi a}{5})+\frac{1}{2}+\frac{1}{2}-\frac{5}{4a\pi}\sin(\frac{2\pi a}{5})=1$. Thus, we can get this result.

九、 Summary and comments：

After completing this experiment, I have learned that MATLAB is a multi-paradigm numerical computing environment that allows for quick and efficient calculations. Some fundamental MATLAB commands for signal representation include: zeros, ones, cos, log, sin, exp, real, imag, abs, angle, linspace, plot, stem, subplot, xlabel, ylabel, title, and more. Additionally, useful commands from the Symbolic Math Toolbox, such as sym, subs, and ezplot, can assist in signal representation and analysis. By plotting signals and systems, we can directly analyze characteristics like periodicity, system properties, trends, time shifts, and scaling.

十、 Suggestion for this lab：

None.

core：

Instructor：Aynul Islam